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An automatic electronic nursing records analysis system based on the text classification and machine learning

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***Abstract*—Enormous amount of unstructured electronic health record are invaluable for the medical research in finding the relationship between the patient's disease and the final diagnosis. How to use computer automatically dig up these information has long been a hot spot. To get the relationship between the clinical outcomes and free text writing by nurse, we developed an automatic categorization system process natural language nursing record based on vector space model. 210 cases of electronic nursing records, which were diagnosed as pancreatitis, were induced in this study. We filtered the restricted corpus for acute pancreatitis classification by information gain (information gain. IG), and construct a text classification system based on Partial least squares discrimination algorithm (PLS-DA) and vector support machine (VSM). PLS loading value analysis found that there are 20 terms can be used to classify medical record text. Our innovative machine-learning algorithm effectively classified free texts of nurse care records associated with normal and acute pancreatitis diagnoses, after training on pre-classified test sets by PLS. This automatic identification technology focus in large-scale medical document may provide important clues to study the acute pancreatitis and other important common disease.**

Keywords-Text classification; Partial Least Squares; Vector support machine; Information gain; Pancreatitis

I. INTRODUCTION

With the rapid development and popularity of the electronic medical record (EMR) technology, a sharp increase of medical record text messages was stored in a readable form by computer. How to automatic classification, organization and management the voluminous literature, information and data (most of it is the text) has become an important topic for the research in text mining and machine learning. Nursing record include a lot of analysis and description of original information from patients. Combined with physical examination and various clinical laboratory test results, the medical record of the original information can often reflect the patient's condition changes in overall situation and display a high correlation with the final clinical diagnosis. Until now, no one has been able to develop software that can automatically identify and understand nursing medical record text. However the development of the software and algorithm for a particular disease process plays an important role in the cognition to the disease [1].

How to classification the severe acute pancreatitis from the normal acute pancreatitis in clinical diagnosis has been the important challenge to a clinician. For clinicians who faced with a patient with abdominal pain, vomiting, fever, hematuresis and amylase rise, which mean he/she could be diagnosed with acute pancreatitis, but a series of imaging examination including ultrasound is key to get the final diagnosis. There has long been a lack of a quantitative research in mining the relationship between the final diagnosis and the early clinical diagnosis. Electronic Nursing records summary is clinical summary recorded by nurse after the first time of nursing ward round, which containing the unbiased digital text of summarized clinical observation. Information contain in this sort of record is likely to be closely related to the final outcome and prognosis of patients. There are generous of nursing record written in Chinese characters, to construct the proper classifier, we need to transfer the large amount of Chinese writing unstructured nursing document into structured documents through auto segmentation by computer. At present, there are two ways transfer the unstructured documents into structured documents, one is Knowledge Engineering (KE) and another is Machine learning (ML) for extraction the key definition[2]. On the extraction in relatively smaller amount of data, the KE technology's extraction effect is better. For unknown large number of the records, the usage of machine learning has more advantages. On one hand, nursing medical record system is composed by a large number of professional terms to describe the short text; on the other hand, there is no document show that it can describe the patient's condition with specific disease accurately through a small number of professional terms. In order to gain the most important professional term from a large number of descriptive sentences, the machine learning is very important method to extract medical record information.

Based on HowNet knowledge base to conversion and structuring the nursing records, We performed the pattern recognition in using the partial least squares and support vector machine (SVM) to find out a way to get specific

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disease information from natural description language algorithm [3].

II. THE THEORETICAL BASIS OF TEXT CLASSIFICATION SYSTEM

Automatic text classification (text categorization), is a method of text mining in pattern recognition and natural language processing technology on many of text automatic identification technology. It is under the predefined system of classification, according to the features of text content (or attributes), will be given a text that is associated with one or more categories of the process. So the text classification can be divided into two parts: understanding of the text content and text pattern classification[3]. Overall, natural language understanding and classification can be understood in order to obtain a function like this: Φ: D * C - > {T, F}, where D is need classification document collections, C according to predefined categories set of classification system. T documents belong to the C, F does not belong to C. That is to say, the final purpose of document classification is to find a mapping function, the realization of accurate D * C to the mapping T or F. The mapping function, namely, we often call it as classifier.To achieve the classifier, we divided the research into the following several parts:

A. Automatic word segmentation and Chinese information extraction

Unstructured information description of patient records for automatic Chinese word segmentation and information extraction, for Chinese language peculiar to a sentence of several word segmentation methods according to the principle of maximum probability participle word segmentation, combined with the overall case language will be one of the biggest probability data string as the segmentation results. Its main principle is as follows

$$P(W|S)=\frac{P(S|W)\times P(W)}{P(S)}\approx P(W)$$

$$P(W)=P(w_1,w_2,\ldots,w_i)\approx P(w_1)\times P(w_2)\times\ldots P(w_i)$$

$$P(w_i)=\frac{\text{The number of times } w_i \text{ appeared in the corpus(n)}}{\text{Total of the times in corpus(N)}}$$

Through an important keyword in the above method to extract the medical record data and the statistics every frequency of key words in the chart.

B. Obtain vector space model

Medical record text is made up of letters and punctuation string, if you want to allow the computer to efficient processing real text, you must find an idealized representation method. By far, the most common to represent the text and discriminant technology is the vector space model. Implement a feature item needs for Chinese word segmentation. Here, we proposed using HowNet and CNKI database, vector space model technology, mainly through the document as some feature term consisting of a vector. The correlation of the same type of documents depends on the degree of repetition of the main features. Given it has a lot of Chinese "的""地""后" those function words. The distinguished ability of these functions is not so big. Therefore, in constructing the feature, we introduce the concept in the Chinese system. HowNet and CNKI pancreatitis related keywords as features to phase in the data table.

Research has shown that using the concept as a defining feature of Chinese word segmentation system can effectively to describe the corresponding concepts. Based on the above concept, we obtained the important technology of using vector space model descriptive concepts do the screening. Finally obtain the vector space model.

C. Partial least squares model and vector support machine pattern recognition

Using partial least squares technique to compress the keyword frequency dimension reduction, find the text associated with final diagnosis of acute pancreatitis is the most combination.

Partial least squares method is a new method of multivariate statistical data analysis; it was put forward in 1983 by S. old and C.A.lbano et al. It has been very clear that the difference between computer pattern recognition and statistical methods of correlation analysis is very obvious. However, the partial least squares rule them in one union. Under the algorithm of PLS, the regression modeling (multiple linear regression), simplify the data structure Principal component analysis (PCA) and correlation analysis between the two groups of variables (canonical correlation analysis) was combined together [4]. In this study, the researchers used partial least squares, establishment and mild pancreatitis, severe acute pancreatitis and death three endings between correlation model, and through the value of projection importance (variable importance for the projection, VIP), to determine the several important medical term which have close relation to the clinical outcome [5]. The support vector machine (SVM) is a main solution to the problem of dualistic pattern classification algorithms. In essence, the basic idea of SVM technology is that finding a decision in the vector space plane; let the plane to the "best" separation of two kinds of data points. If there are more than two kinds of object to classified, support vector machines can process it through establish more than two classes of linear discriminant classifier [6].

D. Performance evaluation of medical record classifier

Recall rate: $r=\frac{a}{a+c}\times 100\%$

Accuracy rate: $p = \frac{a}{a+b} \times 100\%$

F-value: $F_\beta = \frac{(\beta^2+1) \times p \times r}{\beta^2 \times p + r}$

There into, *a* is the system output of the right result number, c is the number of incorrect results in the system output. b is the correct number of the result in test corpus. β is the parameters to adjust the accuracy and recall rate proportion in evaluation function, if consider the recall rate and accuracy are equally important, usually get β = 1, then the evaluation index is

$$F_1 = \frac{2 \times p \times r}{p + r}$$

III. METHOD AND RESULT

A. Basic Work Flow

Text classification includes preprocessing, text representation and classifier recognition, after that, the evaluate classification accuracy is need, so we can draw the whole research process by Fig. 1

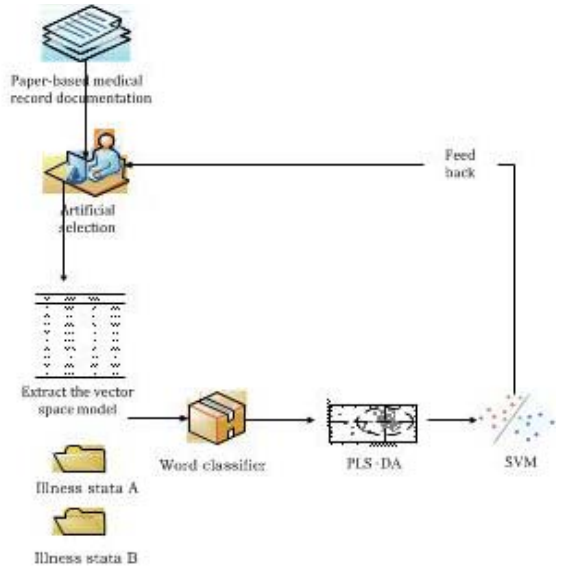


Figure 1. Work flow

Select the 1320 cases records in Sichuan provincial people's hospital that containing the chief complain of abdominal pain in January 2009 - January 2012. All of the chief complained of the nurse record is identified and righteousness segmented by the *core lucene* open source toolkit, and get out of 80 cases of trauma patients [7]. All of the patients medical records information extraction are Obey the S*ichuan people's hospital of medical information resource sharing, security and confidentiality regulations* for management and data extraction, extraction of data does not include the patient's name and address and other personal information.

B. The text feature extraction

Medical record text is made up of letters and punctuation string, if you want to allow the computer to efficient processing real text, you must find an idealized representation method. Until now the most common discriminant technology is the vector space model. Here, we proposed HowNet and CNKI database, vector space model technology, mainly through the document as some feature (feature term) consisting of a vector. The same type of document a correlation between the main features of the degree of repetition. Given there are a lot of Chinese function word such as "的""得""之后" these function word in the medical record is not so important in extracting the substance from he case record, therefore, we introduce the concept from Chinese system such as HowNet and CNKI, the pancreatitis related key words in these system was looked as features words in the extraction.

Based on the above concept, we obtained the technology of using vector space model to descriptive the screening concepts. Finally we get the vector space model. The basic operation flow and result is as follows:

Original:

Chief Complaint：上腹部剧烈疼痛伴呕吐两天。

History of illness：患者于前日下午1时左右出现上腹部疼痛，剑突下为甚。疼痛剧烈难以忍受，呈持续性，放射至腰背部，伴有呕吐，呕吐物为胃内容物。于锦江区妇幼保健院检查血糖依次为28.4mmol/l、26.9mmol/l、33.3mmol/l，血淀粉酶为865U。怀疑为急性重症胰腺炎（SAP），经过对症治疗。患者腹痛缓解，并逐渐出现腹胀。因患者病情较重，于昨日下午2时就诊于我院急诊科。发病以来精神差，未进饮食，无大小便。

vector space model after transformed:

上腹 疼痛 呕吐 剑突 剧烈 忍受 持续 放射 腰背部 胃 成都市 锦江区 医院（保健院） 对症 腹痛 腹胀 病情 急诊 发病 精神 饮食 大便 小便

It must be clear that the conceptual set of the space model basically constitutes the unbiasedness description of the text document feature; the feature space model will change if the medical record were changed. But many of the concepts in the feature space model have little relevance to the difference of acute pancreatitis and ordinary pancreatitis, need to increase the use of partial least squares method to build a supervised learning model to select important feature from the record.

C. The text pattern recognition

Statistics on each copy of the medical records in the table of the word frequency, construction of m * n matrix

X, where m was looked as the number of words, n is the medical record number. Meanwhile, the final diagnosis of discharge was built to matrix Y. using partial least squares discriminant algorithm obtains medical record text and chart, projection variable importance. It must be noted that can characterize medical record content words is not necessarily able to distinguish between acute and common acute pancreatitis.

Justify the clinical outcome and actual clinical outcome by support vector machine to the text and get the SVM cross validation parameters distribution.

IV. RESULT

Vector space model identification showed that all cases of unstructured parts can be converted into 40-50 concepts. Compared with HowNet and CNKI corpus library, in addition to the patient name, there is no other special term outside of the library concept. Information gain method show that if you want to maximum differentiate between medical record information, select 26 basic concept can completely separate 1320 medical records information. (As showed in Table I)

TABLE I. 26 KEY WORDS EXTRACT FROM NURSE RECORDS

Words	Frequency	Weighting	P
反复	0.024491	0.54	0.075567
呕吐	0.025858	0.28	0.048508
纳差	0.035895	0.77	0.04964
疼痛	0.040011	1.21	0.068229
压痛	0.027081	0.35	0.085201
放射	0.051771	0.61	0.025506
……	……	……	……
大便	0.054517	1.55	0.015277
小便	0.017243	1.56	0.047734

Based on partial least squares discriminant analysis (PLS-DA), it can be seen that there is an obvious difference between the severe acute pancreatitis and normal acute pancreatitis cases, according to the medical record of the key word frequency (Fig. 2).

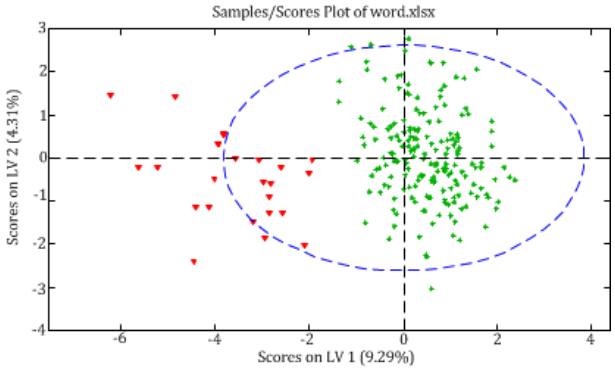


Figure 2. The PCA score plot of the different nurse record, ▼(red) represent the acute pancreatitis, *(green) represent the normal pancreatitis, ellipse represent 95% confidence intervals

We found that the main Chinese Word which can distinguished the severity pancreatitis respectively are

"上腹, 剑突,腰背部, 腹胀, 发病, 饮食" et cl. (Fig.3)

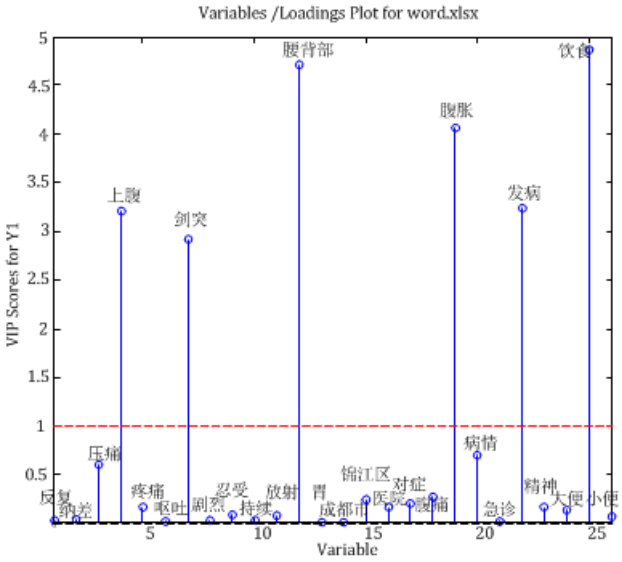


Figure.3 The value of variable importance projection (VIP) for each Chinese word in nurse record

Vector support machine of cross validation shows that the SVM model is the overall classification accuracy is high, the maximum error is only 0.11, the optimum parameters of the classification effect scope by the dark area is given in Fig.4.

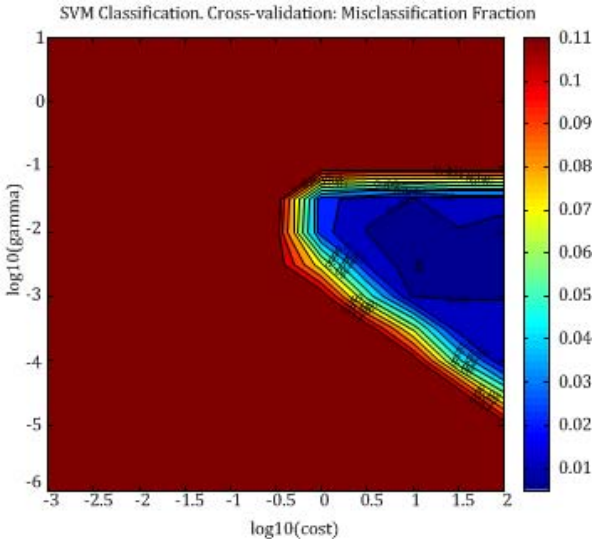


Figure.4 The error rate distribution of support vector machine cross validation

The performance evaluation of medical record classifier's (Table II):

TABLE II. CLASSIFIER PERFORMANCE EVALUATION TABLE

	Severe Pancreatitis	**Pancreatitis**
severe pancreatitis(mark)	22(a)	3(b)
Pancreatitis (mark)	4(c)	181(d)

Thus,

Recall rate： 84.6%

Accuracy rate ：88.0%

F-Value ：0.863

V. DISCUSSION

Structured electronic medical records are the foundation of the machine learning of electronic medical records. But in a structured electronic medical record there are large amounts of unstructured information, such as medical advice and treatment records. A nurse often uses natural language express all kinds of information in the record. But for the computer, to understand these records is difficult. Even the same patients, different nurse using natural language to describe the same symptoms of the expression can be different, but when the whole document was processed computer text mining technology, we can see that the word "上腹疼痛""淀粉酶""腹胀""呕吐""疼痛" et, occur very frequently in acute pancreatitis. These descriptive text phrases may constitute an important difference for severe acute pancreatitis, and this information is likely to contain the nurse intuitive observation for the patient, which involved the pain degree, presence of infection and severity of acute abdominal or other important information. Using the computer to identify and review the information through visits and physical examination plays an important role in early clinical diagnosis and treatment of acute pancreatitis.

Through the medical record of unstructured text data mining, based on partial least squares and support vector machine automatic computer pattern recognition techniques, we found the care of the medical record text writing final diagnosis has a close relationship with the patient. When medical record text appears in the upper abdomen, xiphoid process, lower back and abdominal distension when these terms appeared frequency, the diagnosis of acute pancreatitis could be confirmed in a great extent. PLS show that in all text hyperplane distribution of concept formation, with a pointer to a specific area of the acute pancreatitis. Take this can build out the difference between ordinary text classifier pancreatitis and the acute pancreatitis. By verifying the correctness of the classifier, can obtain classifier the recall rate was 84.62%, the accuracy is 88%.

Generally speaking, a text classification, the accuracy and recall rate contradict each other. If you want to improve the accuracy needs to sacrifice certain recall rate, and vice versa. In many cases, separately considering the accuracy and recall rate to evaluate the classifier is not comprehensive. According to our country "863" 2004 theme of computer software and hardware organization of Chinese information processing and intelligent human-machine interface evaluation task of text, in 100 by the Chinese library classification fourth edition random documents can only be classified evaluation results, the measure value of the highest F system can reach 73.96%. Our system got a better classification result than the study and this is because we adopted by the highly specific nursing medical record text and text classification is highly specific (severe acute pancreatitis, severe acute pancreatitis) are inseparable.

VI. CONCLUSION

Structured electronic medical records are the premise of text mining. Massive amounts of unstructured information of descriptive language could be analysis and processing by computer. We can use computer automatic text analysis process the unstructured nursing medical records by automatic word segmentation, and extract the key term for clinical prognosis. This kind of automatic identification technology focus in large-scale medical document may provide important clues to study for the acute pancreatitis and other important common disease.

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